Python Function Commenter

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Overview:

- The project proposals and viability
  - Proposed project viability
  - The project chosen
- The project's original scope and changes to scope overtime
- Timeline proposed and timeline followed
- Preliminary research on the project
- Methodology chosen vs methodology used
- Accuracy
- Prototype documentation and requirements
- Live Demo of prototype
- Costs
- Conclusion and lessons learned

Project Proposals

- Image to text to translation
- Audio to text to translation
- Text to text translation
- Auto "essay" typer
- Auto report documentation
- Code commenter

Image to Text Translation

- Idea:
  - A user would be able to take an image of non-English text and it will automatically translate it to English and display to the user
- Issues:
  - None of our team members spoke anything fluent besides English
  - Less useful for the team since many viable options exist
- Benefits:
  - A company that cannot use online translations can use this application.

Audio to Text Translation

- Idea:
  - A user would be able to speak to their device in a non-English language and it will automatically translate it to English and display to the user
- Issues:
  - Same issues as Image to Text Translation
- Benefits:
  - A company that cannot use online translations can use this application.

Text to Text Translation

- Idea:
  - A user would be able to input a non-English document and the output will be in English.
- Issues:
  - Same issues as Image to Text Translation and Audio to Text Translation
- Benefits:
  - A company that cannot use online translations can use this application.

Auto "Essay" Typer

- Idea:
  - A user should be able to prompt the application with a short sentence on a topic. The application should then be able to write out a short essay of at least 5 paragraphs.
- Issues:
  - Risk of plagiarism.
  - Difficulty in finding open source texts and essays.
  - Ethical concerns on the application's outputs being considered plagarism if users were to use them.
- Benefits:
  - Creating essays is quick and easy.

Auto Report Documentation

- Idea:
  - A user should be able to input text and output in a documented format.
- Issues:
  - Not a natural language processing project (a requirement of ours).
  - Application not useful due to each company using differing documentation
- Benefits:
  - Having nonstandard documentation be converted to the proper format is a time saver
  - If regulations require a standardized documentation then old nonstandard documents can be easily converted

Code Commenter

- Idea:
  - A user should be able to input code and it should output the code with comments
- Issues:
  - Difficulty in finding accessible non-proprietary code
  - Many different coding languages and versions
- Benefits:
  - Uncommented code can now be commented which can save a lot of time at companies.
  - Useful for our team specifically

The Chosen One

- Code Commenter
  - Why?
    - Useful for us as a tool
    - Is an NLP problem in disguise
    - Non-proprietary code exists online

Code Commenter Scope

- Original Scope:
  - Language: Python 3.1x.x
  - Pytorch
  - OS: Windows
  - Comments will generate for each line of code and be appended to each line respectively.
  - Will utilize CodeBERT model.
- Final Scope:
  - Language: Python 3.1x.x
  - Pytorch
  - OS: Windows
  - Comments will generate only for function definitions and only if the file contains function definitions. Each generated comment will be appended before the function definitions.
  - Will utilizes T5.

Original Timeline

- Milestone 1: Development Environment Setup
  - Date: April 29
  - Activities:
    - Project scope set in place.
    - Install necessary software and tools on Windows 10 or Windows 11 systems.
    - Ensure all team members have Python 3 installed and configured.
- Milestone 2: Prototype Development
  - Date: May 3
  - Activities:
    - Develop initial code to read and split Python 3 code from text files.
    - Begin integrating NLP GAN for generating line-by-line comments.
- Milestone 3: Core Functionality Implementation
  - Date: May 20
  - Activities:
    - Implement functionality to automatically generate comments.
    - Implement error handling for non-commented or pre-commented code inputs.
- Milestone 4: Testing and Feedback Collection
  - Date: May 25
  - Activities:
    - Conduct internal testing of the software.
    - Collect feedback from potential users like fellow students and instructors.
    - Begin making necessary adjustments based on feedback.
- Milestone 5: User Interface Improvement
  - Date: May 29
  - Activities:
    - Develop and refine user interface, possibly adding drag and drop functionality.
    - Enhance user experience based on test feedback.
- Milestone 6: Final Testing and Documentation
  - Date: June 1
  - Activities:
    - Conduct final thorough testing of the software.
    - Prepare user documentation and README files.
    - Ensure all functionalities are working as expected.
- Milestone 7: Project Submission and Presentation
  - Date: June 4
  - Activities:
    - Finalize all project deliverables.
    - Prepare for project presentation and demonstration.
    - Submit the project to the course instructor or project panel.

Actual Timeline

- Milestone 1: Scope
  - Date: April 29 - May 1
  - Activities:
    - Project scope set in place.
- Milestone 2: Setting Up Systems
  - Date: May 1 - May 3
  - Activities:
    - Install necessary software and tools on Windows 10 or Windows 11 systems.
    - Ensure all team members have Python 3 installed and configured.
    - Dataset(s) acquired.
- Milestone 3: Research and Beginning Code
  - Date: May 3 - May 15
  - Activities:
    - Research on GANs and NLP
    - Tokenization completed.
- Milestone 4: Small Scope Change and GUI
  - Date: May 17
  - Activities:
    - A change from CodeBERT to T5
    - Exploration in GUI
- Milestone 5: Fine Tuning Begins
  - Date: May 17 to May 23
  - Activities:
    - Finalized code to enable tuning of T5
- Milestone 6: Fine Tuning Acceptably Done
  - Date: May 23 to May 31
  - Activities:
    - Fine tuning for T5 in acceptable state
    - Initial creation of GUI
- Milestone 7: Project Prototype Completed
  - Date: May 31 to June 3
  - Activities:
    - Final Prototype with GUI
    - ReadMe file completed.
    - Accuracy scores compiled/aggregated.
- Milestone 8: Project Submission and Presentation
  - Date: June 3 to June 7
  - Activities:
    - Documentation completed
    - Presentation

Preliminary Research

- Utilized HuggingFace
  - Data source
    - Used Dataset
  - NLP Research
    - NLP Course
- CodeBERT
  - Intro To Bert
  - CodeBERT Base Data
  - CodeBERT Research Paper
- NLP/ML In General
  - Embedding
  - Transformers
- T5 Code Source
  - Code used as base of application

Data

- T5 Model Code - Fine-tune CodeT5 for generating docstrings from Ruby code.ipynb - Colab (google.com)
- Data Source - espejelomar/code_search_net_python_10000_examples · Datasets at Hugging Face

License

- MIT
- MIT

Methodology Proposed

- Use CodeBERT as our pretrained model and fine tune using a datset found on HuggingFace, Kaggle, Github, or anywhere else that has an acceptable license.
- GUI will utilize either R Shiny, HuggingFace Spaces and Gradio, or Flask and Heroku.
- Each "word" will be tokenized
  - A "word" is defined as characters split by a space on both ends.
    - Must be a minimum of 2 characters.
    - Unique "words" that show up fewer than 10 times will be omitted.
    - English

Methodology Used

- Use T5 as our pretrained model and fine tune using a dataset found on HuggingFace.
- GUI will use the tkinter library in python.
- Each "word" will be tokenized
  - A "word" is defined as characters split by a space on both ends.
    - Must be in English

Difficulties and Helpful Things

- Difficulties
  - Sam went to work.
    - Gone for a week and half the time
  - No DS computer till week 5
  - No teams until week 2ish
  - Gitlab/GitHub setup
  - DS computer being difficult with shared environments
- Helpful Things:
  - Joe

Df <- c('Conclusion', 'Lessons learned', 'Things We Would Do Different')

- Crunch exists
- This was a translation problem the whole time.
- A lot of this stuff exists online already
- Scope creep/changes
- Not use the DS computer. Lots of time wasted.

Accuracy

- A score from 0 to 2 was used to score over 90 function definitions and their generated comments.
  - 0 is bad, 1 is neutral, 2 is good.

Reporter Name	Average Accuracy
Sam Thach	1.186813
Caleb Hart	1.560440
Parker Pratt	1.488889
Josh Aguayo	1.488889
Average Score	**1.431258**

Costs

Name	Wage	Equipment	Total Sum
Sam Thach	40hr x $50 = $2000	$200 Laptop + $3000 PC = $3200	$5200
Caleb Hart	32hr x $50 = $1600	$600 Laptop	$2200
Parker Pratt	30hr x $50 = $1500	$700 Laptop	$2200
Josh Aguayo-Silva	27hr x $50 = $1350	$800 Laptop	$2150
Provided Equipment	NULL	$5000 PC	$5000
Total	**$6,750**	**$10,300**	**$16,750**

Prototype Demonstration and Requirements

APP Demo.mkv

Why it's good/bad.

- Issues:
  - Only works on function definitions.
  - Code needs to be uncommented
- Benefits:
  - Documents undocumented functions
  - Good for large volume of uncommented code

Concluding

Code Commenter

- Idea:
  - A user inputs a txt file containing uncommented python function definitions
- Issues:
  - Code only works on function definitions
  - Code takes into account function names and variable names.
- Benefits:
  - Uncommented code can now be commented which can save a lot of time at companies.
  - Useful for our team specifically

Accuracy

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References

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